CHAPTER 20 · DERIVATIONS

# OSCILLATIONS

3 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 VELOCITY IN SHM

SYLLABUS 20.1(C)

**RESULT** $v = \pm\omega\sqrt{A^2 - x^2}$

1. $x = A\sin\omega t$, so $v = \frac{dx}{dt} = A\omega\cos\omega t$.
2. $\cos\omega t = \pm\sqrt{1 - \sin^2\omega t} = \pm\sqrt{1 - \frac{x^2}{A^2}}$.
3. $\mathbf{v = \pm\omega\sqrt{A^2 - x^2}}$; maximum $\omega A$ at $x = 0$, zero at $x = \pm A$.

## 2 KINETIC AND POTENTIAL ENERGY IN SHM

SYLLABUS 20.2(A)

**RESULT** $K = \frac{1}{2}m\omega^2(A^2 - x^2)$, $U = \frac{1}{2}m\omega^2x^2$

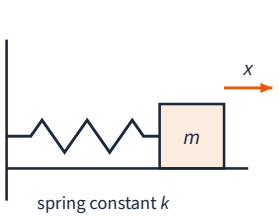
1. $K = \frac{1}{2}mv^2 = \mathbf{\frac{1}{2}m\omega^2(A^2 - x^2)}$.
2. Restoring force $F = -m\omega^2x$. Work done against it from 0 to x: $U = \int m\omega^2x\,dx = \mathbf{\frac{1}{2}m\omega^2x^2}$.
3. Total energy $K + U = \frac{1}{2}m\omega^2A^2$: constant.

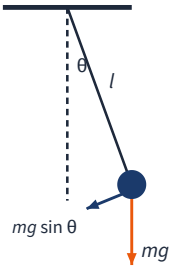
## 3 PERIOD OF A SPRING-MASS SYSTEM AND A SIMPLE PENDULUM

SYLLABUS 20.3(A)

**RESULT** $T = 2\pi\sqrt{\frac{m}{k}}$, $T = 2\pi\sqrt{\frac{l}{g}}$

1. **Spring–mass**: displaced x from equilibrium, restoring force $F = -kx$. $ma = -kx$, so $a = -\frac{k}{m}x$.
2. This is SHM ($a = -\omega^2x$) with $\omega^2 = \frac{k}{m}$: $\mathbf{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}}$.
3. **Simple pendulum**: displaced by angle θ, restoring force along the arc $= -mg\sin\theta \approx -mg\theta$ for small θ, and $\theta = \frac{x}{l}$.
4. $ma = -\frac{mgx}{l}$, so $a = -\frac{g}{l}x$: $\omega^2 = \frac{g}{l}$ and $\mathbf{T = 2\pi\sqrt{\frac{l}{g}}}$.




Valid for small oscillations (θ less than about 10°).

CHAPTER 21 · DERIVATIONS

# WAVE MOTION

3
derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 PHASE DIFFERENCE

SYLLABUS 21.1(D)

**RESULT** $\varphi = kx = \dfrac{2\pi x}{\lambda}$

1. Two points one wavelength apart differ in phase by $2\pi$.
2. Phase difference is proportional to separation: $\dfrac{\varphi}{2\pi} = \dfrac{x}{\lambda}$.
3. $\boldsymbol{\varphi = \dfrac{2\pi x}{\lambda} = kx}$, with $k = \dfrac{2\pi}{\lambda}$.

## 2 WAVE SPEED

SYLLABUS 21.1(E)

**RESULT** $v = f\lambda$

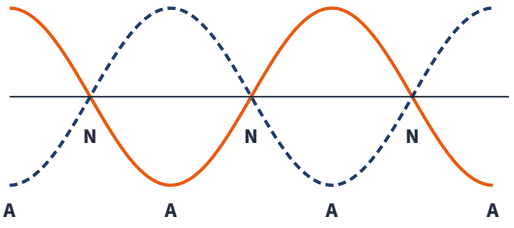
1. In one period T the wave moves forward one wavelength $\lambda$.
2. $v = \dfrac{\lambda}{T}$, and $f = \dfrac{1}{T}$, so $\boldsymbol{v = f\lambda}$ (equivalently $v = \dfrac{\omega}{k}$).

## 3 STANDING WAVE EQUATION

SYLLABUS 21.4(B)

**RESULT** $y = 2A \cos kx \sin \omega t$

1. Two identical waves travel in opposite directions: $y_1 = A \sin(\omega t - kx)$, $y_2 = A \sin(\omega t + kx)$.
2. Superposition: $y = y_1 + y_2$. Use $\sin P + \sin Q = 2 \sin \dfrac{P+Q}{2} \cos \dfrac{P-Q}{2}$.
3. $\boldsymbol{y = 2A \cos kx \sin \omega t}$.
4. Amplitude $2A \cos kx$ depends on position: nodes where $\cos kx = 0$ ($x = \dfrac{\lambda}{4}, \dfrac{3\lambda}{4}, \ldots$), antinodes where $|\cos kx| = 1$ ($x = 0, \dfrac{\lambda}{2}, \lambda, \ldots$). Adjacent nodes are $\dfrac{\lambda}{2}$ apart; no energy is carried along.




N = node, A = antinode; N to N = λ/2

CHAPTER 22 · DERIVATIONS

# SOUND WAVES

2 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 BEAT FREQUENCY

SYLLABUS 22.4(B)

**RESULT** $f = f_1 - f_2$

1. At a point: $y_1 = A \sin 2\pi f_1 t$ and $y_2 = A \sin 2\pi f_2 t$.
2. $y = y_1 + y_2 = 2A \cos(2\pi \frac{f_1 - f_2}{2} t) \sin(2\pi \frac{f_1 + f_2}{2} t)$.
3. The sound has the average frequency $\frac{f_1 + f_2}{2}$ with a slowly varying amplitude $2A \cos(\pi(f_1 - f_2)t)$.
4. Loudness is greatest whenever the amplitude is $\pm 2A$, which happens $(f_1 - f_2)$ times per second: **beat frequency $f = f_1 - f_2$**.

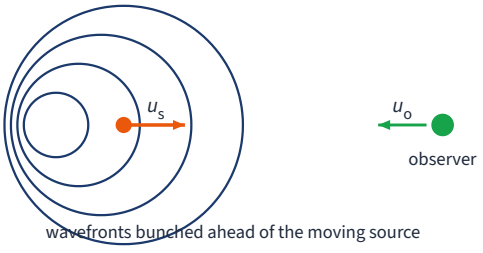
The syllabus notes that this derivation is required.

## 2 DOPPLER EFFECT

SYLLABUS 22.5(B)

**RESULT** $f' = \frac{f(v \pm u_o)}{v \mp u_s}$

1. Source of frequency f; speed of sound v.
2. **Moving source** (speed $u_s$ towards the observer): in one period the source moves $\frac{u_s}{f}$, so the wavelength in front is shortened to $\lambda' = \frac{v - u_s}{f}$.
3. **Moving observer** (speed $u_o$ towards the source): the waves pass the observer at relative speed $v + u_o$.
4. $f' = \frac{\text{relative speed}}{\lambda'} = \mathbf{\frac{f(v + u_o)}{v - u_s}}$ when approaching.
5. When moving apart, change the signs: $f' = \frac{f(v - u_o)}{v + u_s}$.



wavefronts bunched ahead of the moving source

Source and observer move along the same line.

CHAPTER 23 · DERIVATIONS

# GEOMETRICAL OPTICS

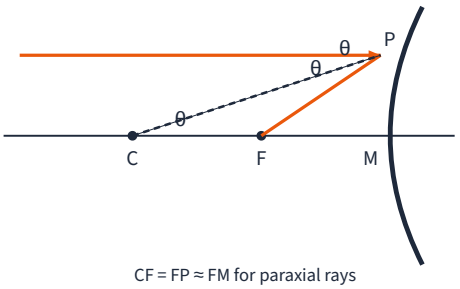
**2** derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 FOCAL LENGTH OF A SPHERICAL MIRROR

SYLLABUS 23.1(A)

**RESULT** $f = \frac{r}{2}$

1. A ray parallel to the axis strikes a concave mirror at P. The normal at P passes through the centre of curvature C.
2. Angle of incidence = angle of reflection = θ. The parallel ray makes angle θ with CP too (alternate angles), so triangle CFP has two equal angles.
3. CF = FP (isosceles triangle).
4. For paraxial rays P is close to the pole M, so FP ≈ FM. Then CM = CF + FM ≈ 2FM.
5. r = 2f, so $\mathbf{f = \frac{r}{2}}$.



CF = FP ≈ FM for paraxial rays

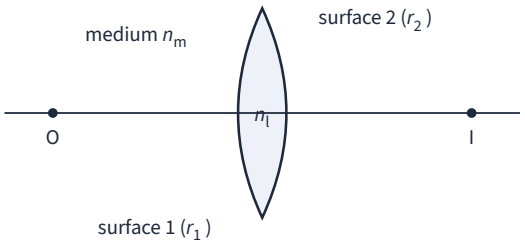
Use the sign convention consistently.

## 2 THIN LENS FORMULA AND LENSMAKER'S EQUATION

SYLLABUS 23.3(A)

**RESULT** $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}; \frac{1}{f_m} = (\frac{n_l}{n_m} - 1)(\frac{1}{r_1} - \frac{1}{r_2})$

1. Lens of refractive index $n_l$ in a medium $n_m$. Use $\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$ at each surface.
2. Surface 1 (medium → lens): $\frac{n_m}{u} + \frac{n_l}{v_1} = \frac{n_l - n_m}{r_1}$ … (1)
3. The image $I_1$ is the object for surface 2. The lens is thin, so it is a virtual object at $u_2 = -v_1$.
4. Surface 2 (lens → medium): $-\frac{n_l}{v_1} + \frac{n_m}{v} = \frac{n_m - n_l}{r_2}$ … (2)
5. Add (1) and (2): $\frac{n_m}{u} + \frac{n_m}{v} = (n_l - n_m)(\frac{1}{r_1} - \frac{1}{r_2})$.
6. Divide by $n_m$: $\frac{1}{u} + \frac{1}{v} = (\frac{n_l}{n_m} - 1)(\frac{1}{r_1} - \frac{1}{r_2})$.
7. Object at infinity: $v = f_m$, so $\mathbf{\frac{1}{f_m} = (\frac{n_l}{n_m} - 1)(\frac{1}{r_1} - \frac{1}{r_2})}$ (lensmaker's equation), and therefore $\mathbf{\frac{1}{u} + \frac{1}{v} = \frac{1}{f}}$.




In this form r is positive when the centre of curvature is on the side the light goes to, so for a biconvex lens $r_2$ is negative. Many books write $(\frac{1}{r_1} + \frac{1}{r_2})$ with both radii of a convex lens positive; both give the same f.

CHAPTER 24 · DERIVATIONS

# WAVE OPTICS

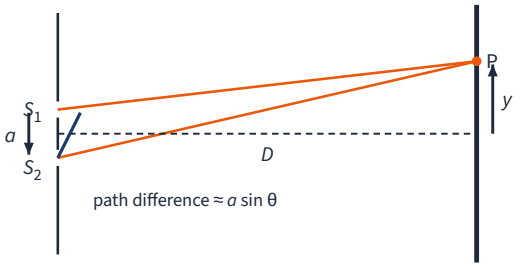
4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 FRINGE SEPARATION IN YOUNG'S EXPERIMENT

SYLLABUS 24.3(B)

**RESULT** $x = \frac{\lambda D}{a}$

1. Slits $S_1$ and $S_2$ a apart; screen at distance D ($D \gg a$). Point P is at height y above the centre.
2. Path difference $S_2P - S_1P \approx a \sin\theta \approx a \tan\theta = \frac{ay}{D}$ (small θ).
3. Bright fringe: $\frac{ay}{D} = m\lambda$, so $y_m = \frac{m\lambda D}{a}$.
4. Fringe separation $\mathbf{x = y_{m+1} - y_m = \frac{\lambda D}{a}}$.

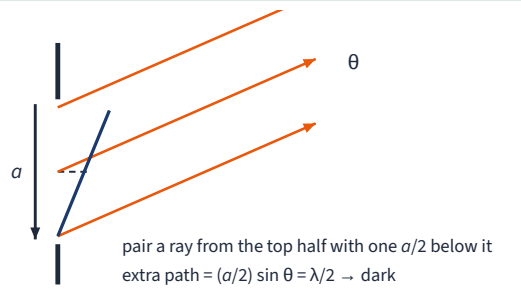



## 2 FIRST MINIMUM OF A SINGLE SLIT

SYLLABUS 24.5(B)

**RESULT** $\sin\theta = \frac{\lambda}{a}$

1. Divide the slit of width a into two halves.
2. Every point in the top half pairs with a point $\frac{a}{2}$ below it. In direction θ the path difference of each pair is $\frac{a}{2}\sin\theta$.
3. If $\frac{a}{2}\sin\theta = \frac{\lambda}{2}$, each pair cancels and the screen is dark.
4. $\mathbf{\sin\theta = \frac{\lambda}{a}}$ for the first minimum.




## 3 DIFFRACTION GRATING

SYLLABUS 24.6(B)

**RESULT** $d \sin\theta = m\lambda$

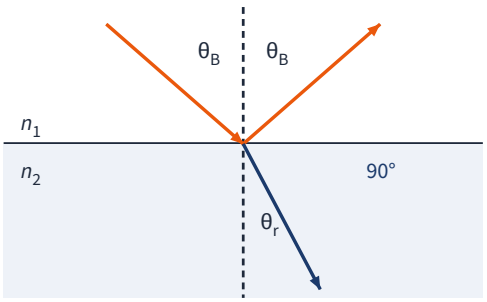
1. Grating spacing $d = \frac{1}{N}$ (N lines per metre).
2. In direction θ, light from adjacent slits has path difference d sin θ.
3. Bright maxima when all slits are in phase: $\mathbf{d \sin\theta = m\lambda}$ (m = 0, 1, 2, …), with m less than $\frac{d}{\lambda}$.

## 4 BREWSTER'S LAW

SYLLABUS 24.7(F)

**RESULT** $\tan \theta_B = \frac{n_2}{n_1}$

1. At the Brewster angle the reflected light is completely plane polarised and the reflected and refracted rays are at 90°.
2. $\theta_B + \theta_r = 90°$.
3. Snell's law: $n_1 \sin \theta_B = n_2 \sin \theta_r = n_2 \sin(90° - \theta_B) = n_2 \cos \theta_B$.
4. $\mathbf{\tan \theta_B = \frac{n_2}{n_1}}$.



CHAPTER 25 · DERIVATIONS

# QUANTUM PHYSICS

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 RADII OF BOHR ORBITS

SYLLABUS 25.3(B)

**RESULT** $r_n = \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}$

1. The electric attraction provides the centripetal force: $\dfrac{e^2}{4\pi\varepsilon_0 r^2} = \dfrac{mv^2}{r}$ … (1)
2. Bohr's postulate (quantised angular momentum): $mvr = \dfrac{nh}{2\pi}$, so $v = \dfrac{nh}{2\pi mr}$ … (2)
3. Substitute (2) into (1): $\dfrac{e^2}{4\pi\varepsilon_0 r} = mv^2 = \dfrac{n^2h^2}{4\pi^2 mr^2}$.
4. $\mathbf{r_n = \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}}$; $r_1 = 5.3 \times 10^{-11}$ m.

## 2 ENERGY LEVELS OF HYDROGEN

SYLLABUS 25.3(C)

**RESULT** $E_n = -\dfrac{me^4}{8\varepsilon_0^2 h^2 n^2}$

1. From (1): $KE = \tfrac{1}{2}mv^2 = \dfrac{e^2}{8\pi\varepsilon_0 r}$.
2. $PE = -\dfrac{e^2}{4\pi\varepsilon_0 r}$.
3. Total $E = KE + PE = -\dfrac{e^2}{8\pi\varepsilon_0 r}$.
4. Substitute $r_n$: $\mathbf{E_n = -\dfrac{me^4}{8\varepsilon_0^2 h^2 n^2}} = -\dfrac{13.6}{n^2}$ eV.

## 3 MINIMUM X-RAY WAVELENGTH

SYLLABUS 25.4(C)

**RESULT** $\lambda_{min} = \dfrac{hc}{eV}$

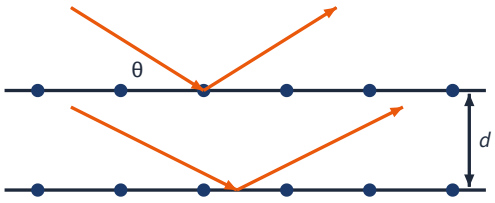
1. Electrons accelerated through p.d. V gain kinetic energy eV.
2. The shortest wavelength is emitted when all of this energy goes into one photon: $eV = hf_{max} = \dfrac{hc}{\lambda_{min}}$.
3. $\mathbf{\lambda_{min} = \dfrac{hc}{eV}}$ (independent of the target metal).

## 4 BRAGG'S LAW

SYLLABUS 25.4(E)

**RESULT** $2d \sin \theta = m\lambda$

1. X-rays are reflected by two adjacent atomic planes a distance d apart, at glancing angle θ.
2. The ray from the lower plane travels an extra path 2d sin θ.
3. Constructive interference when the extra path is a whole number of wavelengths: **$2d \sin \theta = m\lambda$**.




extra path of lower ray = $2d \sin \theta$

CHAPTER 26 · DERIVATIONS

# NUCLEAR PHYSICS

2
derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 RADIOACTIVE DECAY LAW

SYLLABUS 26.2(F)

**RESULT** $N = N_0 e^{-\lambda t}$

1. Decay law: $\frac{dN}{dt} = -\lambda N$.
2. Separate: $\frac{dN}{N} = -\lambda\, dt$.
3. Integrate from $N_0$ at $t = 0$: $\ln \frac{N}{N_0} = -\lambda t$.
4. **$N = N_0 e^{-\lambda t}$.**

## 2 HALF-LIFE

SYLLABUS 26.2(G)

**RESULT** $t_{½} = \frac{\ln 2}{\lambda}$

1. At $t = t_{½}$, $N = \frac{N_0}{2}$.
2. $\frac{N_0}{2} = N_0 e^{-\lambda t_{½}}$, so $e^{\lambda t_{½}} = 2$.
3. $\lambda t_{½} = \ln 2$, so **$t_{½} = \frac{\ln 2}{\lambda}$** $\approx \frac{0.693}{\lambda}$.